

ROMORANTIN-PRUNIER AP, France

WMO: 072470

Lat: 47.3197N

Lon: 1.6875E

Elev: 282

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.0	23.5	14.6	11.7	24.0	19.1	14.5	26.6	21.0	51.9	19.0	51.1	1.7	30	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	90.9	69.3	86.7	67.8	82.9	66.7	71.5	86.2	69.8	83.1	68.1	79.8	6.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	100.6	76.3	65.4	94.8	74.2	63.8	89.7	72.3	35.3	86.2	33.9	83.3	32.5	79.8	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.4	13.4	13.3	97.5	5.7	4.7	9.2	100.9	5.9	103.6	2.6	106.2	-1.5	109.6
			13.3	74.9	5.8	2.3	9.1	76.5	5.7	77.8	2.4	79.1	-1.8	80.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	39.8	40.4	45.7	50.5	56.5	63.8	66.6	65.9	59.5	53.5	45.7	40.4	(6)
(7)		DBStd	11.61	7.43	7.17	6.39	5.81	6.00	6.31	5.46	5.76	5.51	6.88	6.56	7.43	(7)
(8)		HDD50	1340	322	274	163	63	10	0	0	0	1	44	158	305	(8)
(9)		HDD65	4850	781	690	599	435	271	96	45	57	179	357	578	762	(9)
(10)		CDD50	2227	6	4	29	78	212	414	514	492	286	154	30	8	(10)
(11)		CDD65	261	0	0	0	0	7	60	94	84	14	2	0	0	(11)
(12)		CDH74	4114	0	0	1	53	205	885	1344	1264	324	38	0	0	(12)
(13)		CDH80	1488	0	0	0	3	36	324	519	524	79	3	0	0	(13)
(14)	Wind	WSAvg	5.4	6.0	6.3	6.2	5.8	5.4	5.2	5.1	4.4	4.3	4.6	5.6	6.1	(14)
(15)	Precipitation	PrecAvg	27.8	2.3	2.0	1.8	2.4	2.6	2.0	2.3	2.0	2.2	2.7	2.8	2.6	(15)
(16)		PrecMax	35.7	5.2	3.5	5.6	6.2	6.9	3.9	4.6	4.2	5.4	5.2	5.5	5.8	(16)
(17)		PrecMin	20.7	0.2	0.2	0.2	0.3	0.8	0.3	0.3	0.3	0.5	1.2	0.9	0.6	(17)
(18)		PrecStd	4.0	1.2	1.0	1.3	1.3	1.2	1.0	1.0	1.2	1.3	1.1	1.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	63.7	72.0	79.2	84.4	94.8	95.8	98.2	87.3	78.7	65.2	58.6	(19)
(20)		MCWB	52.9	51.3	54.2	60.6	67.6	71.7	69.1	71.2	65.3	64.3	58.0	53.0	(20)	
(21)		2%	DB	55.0	57.6	67.0	74.5	79.5	87.8	90.1	91.2	82.1	73.4	61.2	56.1	(21)
(22)		MCWB	51.8	49.8	52.7	58.4	63.3	68.7	69.6	69.0	64.7	62.6	56.6	52.7	(22)	
(23)		5%	DB	53.1	54.1	62.5	69.4	75.5	83.4	85.8	86.1	77.8	68.3	58.8	54.1	(23)
(24)		MCWB	50.2	49.0	51.6	56.1	61.9	67.6	67.7	67.6	63.5	60.2	55.1	51.2	(24)	
(25)		10%	DB	51.0	51.8	58.6	65.0	71.3	78.9	81.7	81.1	73.6	64.7	56.2	52.0	(25)
(26)		MCWB	48.2	48.1	50.5	53.9	59.5	65.5	66.7	66.0	61.7	58.6	52.8	49.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	53.6	57.2	63.3	69.2	74.4	72.8	73.1	69.8	65.8	60.0	55.5	(27)
(28)		MADB	56.0	59.0	65.1	76.8	80.7	87.8	86.9	89.7	80.7	75.0	62.5	57.4	(28)	
(29)		2%	WB	52.5	51.5	54.8	59.7	65.6	71.2	71.1	71.0	66.8	63.6	57.3	53.4	(29)
(30)		MADB	54.5	55.0	62.0	71.3	76.3	83.8	85.9	86.7	77.5	70.3	60.8	55.5	(30)	
(31)		5%	WB	50.4	50.0	53.1	57.1	62.9	68.9	69.4	69.0	64.9	61.4	55.2	51.5	(31)
(32)		MADB	52.8	53.5	59.5	66.8	72.7	80.2	82.6	82.5	75.0	67.5	58.4	53.7	(32)	
(33)		10%	WB	48.4	48.3	51.5	54.6	60.6	66.6	67.5	67.2	63.0	59.4	53.2	49.7	(33)
(34)		MADB	50.7	51.7	57.8	63.1	69.6	76.8	79.4	78.2	71.2	63.8	55.9	51.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.6	16.4	21.4	23.4	23.0	23.5	24.5	26.1	25.5	20.2	14.8	13.0	(35)	
(36)		5% DB	MADB	12.5	21.6	31.1	33.7	31.0	32.0	33.6	36.2	33.2	25.3	17.5	14.9	(36)
(37)		MCWBR	9.8	15.0	18.4	18.6	16.5	15.1	14.7	15.4	17.2	15.5	12.8	11.9	(37)	
(38)		5% WB	MADB	11.3	17.1	23.2	29.5	27.3	27.9	30.2	30.8	26.8	22.2	16.4	13.0	(38)
(39)		MCWBR	9.5	12.7	14.7	17.0	15.5	14.1	13.9	14.7	14.9	14.1	12.5	11.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.343	0.372	0.401	0.404	0.416	0.410	0.399	0.368	0.364	0.336	0.320	(40)	
(41)		taud	2.445	2.389	2.314	2.232	2.274	2.276	2.290	2.336	2.404	2.422	2.447	2.429	(41)	
(42)		Ebn at Noon	240	259	268	270	273	270	269	268	266	248	233	226	(42)	
(43)		Edh at Noon	22	28	35	41	41	41	40	37	31	27	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	331	595	982	1408	1656	1870	1812	1594	1235	742	401	284	(44)	
(45)		RadStd	35	83	120	171	152	180	118	146	88	62	43	43	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page